

Use Case Name:	Manage Library Catalog (including View Library Catalog)													
Scenario:	Librarian maintains the catalog database, while Students and Faculty search, view, and select library resources.													
Triggering Event:	Student/Faculty accesses the catalog search interface to find materials, or a Librarian accesses the dashboard to manage inventory.													
Brief Description:	This use case encompasses catalog searching, filtering, and viewing item availability by students/faculty (View Library Catalog), as well as administrative CRUD operations (add, edit, delete) performed by the Librarian to maintain inventory accuracy. It interfaces with Koha ILS and DocuPulse.													
Actors:	Student/Faculty, Librarian, Koha ILS (External System), DocuPulse (External System)													
Related Use Cases:	Includes: View Library Catalog. Extends: Manage Borrowed Books.													
Stakeholders:	• Student/Faculty: To easily find and view required resources. • Librarian: To maintain accurate library records. • DocuPulse: To synchronize external metadata.													
Preconditions:	• The student/faculty successfully signs in to the LIRA system, OR the Librarian successfully signs in to Koha. • The catalog database and LIRA portal are online.													
Postconditions:	• Matching catalog items and availability details are displayed to the user. • The catalog database is successfully updated, saved, and synchronized with Koha ILS and DocuPulse.													
Flow of Activities:	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Student/Faculty successfully signs in to the LIRA system and accesses the catalog search interface.</td><td>1.1 System validates session and loads search interface.</td></tr><tr><td>2. Student/Faculty enters search criteria (e.g., title, author, keyword) and submits.</td><td>2.1 Queries local database records and Koha ILS to retrieve matching items. 2.2 Displays a list of matching catalog items.</td></tr><tr><td>3. Student/Faculty selects a specific item from search results.</td><td>3.1 Displays detailed item information, availability, and location.</td></tr><tr><td>4. Librarian successfully signs in to Koha and accesses the catalog management dashboard.</td><td>4.1 Displays the current catalog list and data entry forms.</td></tr><tr><td>5. Librarian submits new item details, edits existing records, or confirms item deletion.</td><td>5.1 Validates submitted data, updates catalog database, and synchronizes with Koha ILS. 5.2 Displays a success confirmation message.</td></tr></table>		Actor	System	1. Student/Faculty successfully signs in to the LIRA system and accesses the catalog search interface.	1.1 System validates session and loads search interface.	2. Student/Faculty enters search criteria (e.g., title, author, keyword) and submits.	2.1 Queries local database records and Koha ILS to retrieve matching items. 2.2 Displays a list of matching catalog items.	3. Student/Faculty selects a specific item from search results.	3.1 Displays detailed item information, availability, and location.	4. Librarian successfully signs in to Koha and accesses the catalog management dashboard.	4.1 Displays the current catalog list and data entry forms.	5. Librarian submits new item details, edits existing records, or confirms item deletion.	5.1 Validates submitted data, updates catalog database, and synchronizes with Koha ILS. 5.2 Displays a success confirmation message.
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Exception Conditions:	2.1 If no items match the user's search criteria, then the system displays a 'No results found' message. 5.1 If data validation fails during an update (e.g., missing ISBN), then the system highlights errors and prompts the Librarian to correct them. 5.2 If an item being modified or deleted is currently checked out, then the system invokes the Manage Borrowed Books process to resolve status conflicts.													

Use Case Name:	Manage Borrowed Books (including Borrow Books)
Scenario:	Students and Faculty initiate borrow requests, while the Librarian manages inventory lifecycle circulation and status tracking.
Triggering Event:	User initiates a borrow request while viewing a catalog item, or a Librarian processes check-ins, renewals, or returns.
Brief Description:	Handles both user-facing book borrowing requests (Borrow Books) and administrative circulation management (Manage Borrowed Books). It records transactions by communicating with the Koha ILS to manage loan statuses, due dates, and inventory lifecycle.
Actors:	Student/Faculty, Librarian, Koha ILS (External System)
Related Use Cases:	Extension of: Manage Library Catalog. Includes: Borrow Books.
Stakeholders:	• Student/Faculty: To successfully acquire and borrow books. • Librarian & Koha ILS: To track inventory lifecycle, checkouts, and circulation.

Preconditions:	• The student/faculty successfully signs in to the LIRA system with a valid account, OR the Librarian successfully signs in to Koha. • The selected item is available and LIRA is connected to the Koha API.									
Postconditions:	• The item status is updated to 'Checked Out' or returned, the user account reflects the loan, and the transaction is logged in Koha ILS.									
Flow of Activities:	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Student/Faculty successfully signs in to the LIRA system and initiates a borrow request for an available item.</td><td>1.1 Verifies the user's account status and current borrowing limits. 1.2 Prompts the user for final confirmation.</td></tr><tr><td>2. Student/Faculty confirms the borrow request and accepts terms.</td><td>2.1 Transmits transaction data to Koha ILS to log checkout. 2.2 Updates local item status to 'Checked Out' and displays borrowing confirmation receipt and due date.</td></tr><tr><td>3. Librarian successfully signs in to Koha and accesses the circulation module.</td><td>3.1 Queries Koha and displays synchronized status of borrowed books. 3.2 Processes check-ins, renewals, or lost book records and commits updates to Koha.</td></tr></table>		Actor	System	1. Student/Faculty successfully signs in to the LIRA system and initiates a borrow request for an available item.	1.1 Verifies the user's account status and current borrowing limits. 1.2 Prompts the user for final confirmation.	2. Student/Faculty confirms the borrow request and accepts terms.	2.1 Transmits transaction data to Koha ILS to log checkout. 2.2 Updates local item status to 'Checked Out' and displays borrowing confirmation receipt and due date.	3. Librarian successfully signs in to Koha and accesses the circulation module.	3.1 Queries Koha and displays synchronized status of borrowed books. 3.2 Processes check-ins, renewals, or lost book records and commits updates to Koha.
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Exception Conditions:	1.1 If the user has a blocked account or overdue items, then the system denies the borrow request and displays a warning. 2.1 If the item becomes unavailable during the transaction, then the system cancels the process and notifies the user. 2.2 If Koha ILS is unreachable, then the system queues the transaction locally and alerts the Librarian.									

Use Case Name:	Manage AI Chatbot (including Ask AI Chatbot)											
Scenario:	Students and Faculty ask AI questions for immediate research assistance, while the Librarian with admin privileges configures the AI knowledge base and parameters.											
Triggering Event:	User submits a query to the chat widget, or a Librarian accesses the AI configuration panel.											
Brief Description:	Encompasses both automated response generation for student research queries (Ask AI Chatbot) and administrative tuning of chatbot parameters, custom prompts, and knowledge sources by the Librarian (Manage Chatbot).											
Actors:	Student/Faculty, Librarian, AI Engine											
Related Use Cases:	Includes: Ask AI Chatbot.											
Stakeholders:	• Student/Faculty: For immediate, accurate research assistance. • Librarian: To audit AI interactions and ensure response safety and accuracy.											
Preconditions:	• The student/faculty successfully signs in to the LIRA system, OR the Librarian successfully signs in with admin privileges. • The AI engine and RAG pipeline are online.											
Postconditions:	• A relevant research response is generated and displayed to the user, OR the chatbot's configuration and knowledge base are successfully updated.											
Flow of Activities:	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Student/Faculty successfully signs in to the LIRA system and submits a question into the chat widget.</td><td>1.1 Parses the query and routes it to the AI Engine.</td></tr><tr><td>2. AI Engine generates a synthesized response based on RAG prompt context.</td><td>2.1 Formats the retrieved answer, displays it in the chat interface, and logs the interaction.</td></tr><tr><td>3. Librarian successfully signs in with admin privileges and accesses the chatbot configuration panel.</td><td>3.1 Displays current knowledge sources, prompts, and interaction logs.</td></tr><tr><td>4. Librarian uploads new knowledge documents, edits custom prompts, or deletes obsolete knowledge documents.</td><td>4.1 Validates and applies new configurations to the AI engine, or removes deleted sources from the context.</td></tr></table>		Actor	System	1. Student/Faculty successfully signs in to the LIRA system and submits a question into the chat widget.	1.1 Parses the query and routes it to the AI Engine.	2. AI Engine generates a synthesized response based on RAG prompt context.	2.1 Formats the retrieved answer, displays it in the chat interface, and logs the interaction.	3. Librarian successfully signs in with admin privileges and accesses the chatbot configuration panel.	3.1 Displays current knowledge sources, prompts, and interaction logs.	4. Librarian uploads new knowledge documents, edits custom prompts, or deletes obsolete knowledge documents.	4.1 Validates and applies new configurations to the AI engine, or removes deleted sources from the context.
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Exception Conditions:	1.1 If the AI service times out, then the system apologizes and provides contact information for a human librarian. 2.1 If the AI engine cannot understand the query, then the system displays a fallback message asking the user to rephrase. 4.1 If the Librarian uploads an incompatible document, then the system rejects it and displays an error message.
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Use Case Name:	Manage Reports											
Scenario:	The Librarian generates analytics regarding library usage, circulation data, and chatbot interactions.											
Triggering Event:	Librarian requests a statistical or operational report from the dashboard.											
Brief Description:	Allows the Librarian to compile, preview, adjust, and export data reports by extracting data from local LIRA databases and integrating circulation data from the Koha ILS.											
Actors:	Librarian, Koha ILS (External System)											
Related Use Cases:	N/A											
Stakeholders:	• Library Management: To make data-driven decisions regarding acquisitions and system usage.											
Preconditions:	• The Librarian successfully signs in to Koha and holds administrative reporting privileges. • Log data exists in both LIRA and Koha databases.											
Postconditions:	• A formatted report is generated, previewed on the dashboard, and available for download (PDF/Excel).											
Flow of Activities:	<table><tr><th>Actor</th><th>System</th></tr><tr><td>1. Librarian successfully signs in to Koha and selects a report type and date range from the dashboard.</td><td>1.1 Requests necessary usage data from local database and circulation data from Koha ILS.</td></tr><tr><td>2. Koha ILS provides requested raw circulation data.</td><td>2.1 Aggregates combined data and displays report preview on the dashboard.</td></tr><tr><td>3. Librarian adjusts parameters or clicks to export the final report.</td><td>3.1 Generates a downloadable file (PDF/Excel) and provides it to the user.</td></tr><tr><td>4. Librarian dismisses, clears, or deletes an active report preview.</td><td>4.1 Removes the preview data from the dashboard.</td></tr></table>		Actor	System	1. Librarian successfully signs in to Koha and selects a report type and date range from the dashboard.	1.1 Requests necessary usage data from local database and circulation data from Koha ILS.	2. Koha ILS provides requested raw circulation data.	2.1 Aggregates combined data and displays report preview on the dashboard.	3. Librarian adjusts parameters or clicks to export the final report.	3.1 Generates a downloadable file (PDF/Excel) and provides it to the user.	4. Librarian dismisses, clears, or deletes an active report preview.	4.1 Removes the preview data from the dashboard.
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Exception Conditions:	1.1 If Koha ILS is offline, then the system generates a partial report using only local LIRA data and appends a warning note. 2.1 If the selected date range contains no data, then the system outputs an empty report accompanied by a notification.											